

Module - 5

Wave Equations in Transmission Line

Egvt def of a wire txmn line consists of following elements

1) Series Resistance (R)

It represents resistance of txmn line & it indicates ohmic loss in the line. unit Ω/km .

2) Series Inductance (L)

current carrying txmn line has an associated magnetic field which varies with time. It is indicated by series inductance L . Unit is Henry/km.

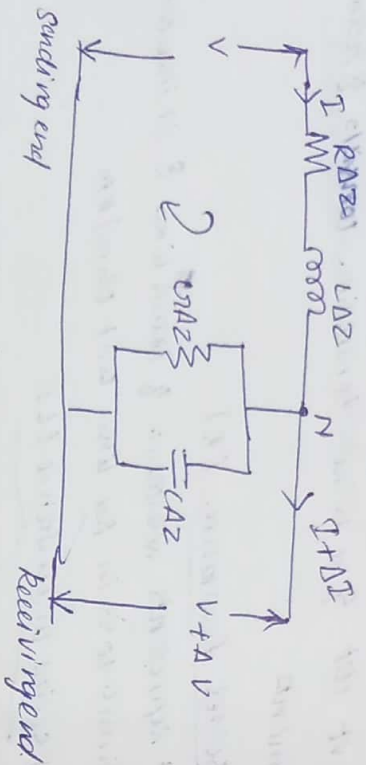
3) Shunt Capacitance (C)

Two conductors separated by a distance in a dielectric medium give rise to capacitance which acts in $\parallel$ with line. Unit Farad/km.

4) Shunt leakage conductance (G)

dielectric b/w the conductors may not be perfect dielectric. A leakage current can flow from one conductor to other conductor through the dielectric. To account for this leakage current, leakage conductance is considered. unit mho/km .

Considering differential length Δz of transmission line
eqn. is shown.



Applying KVL at one loop.

$$V - IR\Delta z - j\omega L\Delta z \frac{\partial I}{\partial t} - [V + \Delta V] = 0$$

$$-IR\Delta z - j\omega L\Delta z \frac{\partial I}{\partial t} - \Delta V = 0$$

$$V = L \frac{\partial I}{\partial t}$$

$$-\Delta V = IR\Delta z + j\omega L\Delta z \frac{\partial I}{\partial t}$$

$$-\Delta V = [IR + j\omega L \frac{\partial I}{\partial t}] \Delta z$$

$$-\frac{\Delta V}{\Delta z} = [IR + j\omega L \frac{\partial I}{\partial t}]$$

Limiting case $\Delta z \rightarrow 0$.

$$\lim_{\Delta z \rightarrow 0} -\frac{\Delta V}{\Delta z} = IR + j\omega L \frac{\partial I}{\partial t}$$

$$-\frac{\partial V}{\partial z} = IR + j\omega L \frac{\partial I}{\partial t}$$

Replacing $\frac{\partial}{\partial t}$ by $j\omega$

$$-\frac{\partial V}{\partial z} = [R + j\omega L] I$$

$$i = c \frac{\partial V}{\partial t}$$

Applying KCL at node N.

$$-I + \underbrace{(V + \Delta V)G\Delta z + C\Delta z \frac{\partial (V + \Delta V)}{\partial t}}_{\text{Veg. x conduct. current}} + I + \Delta I = 0$$

$$-\Delta I = \{ (V + \Delta V)G + C \frac{\partial (V + \Delta V)}{\partial t} \} \Delta z$$

$$-\frac{\Delta I}{\Delta z} = (V + \Delta V)G + C \frac{\partial (V + \Delta V)}{\partial t}$$

As limiting case, $\Delta z \rightarrow 0$, $\Delta V \rightarrow 0$.

$$\lim_{\Delta z \rightarrow 0} -\frac{\Delta I}{\Delta z} = (V + 0)G + C \frac{\partial (V + 0)}{\partial t}$$

$$-\frac{\partial I}{\partial z} = VG + C \frac{\partial V}{\partial t}$$

Replacing $\frac{\partial}{\partial t}$ by $j\omega$

$$-\frac{\partial I}{\partial z} = (VG + j\omega C)V \quad \text{--- (2)}$$

Equations (1) & (2) are called telegrapher's equations

$$\left\{ \begin{aligned} \frac{\partial V}{\partial z} &= -(R + j\omega L)I \\ \frac{\partial I}{\partial z} &= -(G + j\omega C)V \end{aligned} \right\}$$

Differentiating (1) wrt z

$$\frac{\partial^2 V}{\partial z^2} = -(R+j\omega L) \frac{\partial I}{\partial z}$$

Substituting for $\frac{\partial I}{\partial z}$ from eqn (2)

$$\frac{\partial^2 V}{\partial z^2} = -(R+j\omega L) \frac{\partial}{\partial z} (G+j\omega C)V$$

$$\frac{\partial^2 V}{\partial z^2} = (R+j\omega L)(G+j\omega C)V \quad (3)$$

$$\frac{\partial^2 V}{\partial z^2} = \gamma^2 V \quad (4)$$

where $\gamma^2 = (R+j\omega L)(G+j\omega C)$

Similarly differentiate (2) wrt z

$$\frac{\partial^2 I}{\partial z^2} = -(G+j\omega C) \frac{\partial V}{\partial z}$$

Substituting for $\frac{\partial V}{\partial z}$ from eqn (1)

$$\frac{\partial^2 I}{\partial z^2} = -(G+j\omega C) \frac{\partial}{\partial z} (R+j\omega L)I$$

$$\frac{\partial^2 I}{\partial z^2} = (G+j\omega C)(R+j\omega L)I \quad (5)$$

$$\text{Again } \frac{\partial^2 I}{\partial z^2} = \gamma^2 I \quad (6)$$

where $\gamma^2 = (R+j\omega L)(G+j\omega C)$

$$\text{ie) } \frac{\partial^2 V}{\partial z^2} = \gamma^2 V$$

→ transmission eqn.

γ is the propagation constant.

$$\gamma = \sqrt{(R+j\omega L)(G+j\omega C)}$$

General solution of transmission line eqns are

$$V(z) = \underbrace{V^+ e^{-\gamma z}}_{\text{incident wave}} + \underbrace{V^- e^{+\gamma z}}_{\text{reflected wave}}$$

$V^+ e^{-\gamma z} \rightarrow$ wave that propagates along the +ve z direction & $V^- e^{+\gamma z}$ is the wave propagates along -ve z direction.

characteristic impedance (Z_0) (positive real number)

Ratio of V & I at any distance for an infinitely long line is independent of the distance & is called the characteristic impedance.

$$Z_0 = \sqrt{\frac{R+j\omega L}{G+j\omega C}}$$

characteristic impedance depends on R, L, G, C and not on the length of the line, i.e. constant

Lossless transmission line

Propagation constant $\gamma = \sqrt{(R+j\omega L)(G+j\omega C)}$

for lossless line $R=0, G=0$

$$\gamma = \sqrt{j\omega L \cdot j\omega C} = j\omega \sqrt{LC}$$

we know

$$\gamma = \alpha + j\beta$$

$$\Rightarrow \alpha + j\beta = j\omega \sqrt{LC}$$

Equating real & imag. parts $\alpha = 0$

$$\beta = \omega \sqrt{LC}$$

characteristic impedance $Z_0 = \sqrt{\frac{R+j\omega L}{G+j\omega C}}$

for lossless line

$$Z_0 = \sqrt{\frac{j\omega L}{j\omega C}} = \sqrt{\frac{L}{C}} \rightarrow \text{no imag. part}$$

Phase velocity for a lossless transmission line

$$V_p = \frac{1}{\sqrt{LC}}$$

for a lossless transmission line

propagation const. $\gamma = j\omega \sqrt{LC}$

Attenuation const. $\alpha = 0$

Phase const. $\beta = \omega \sqrt{LC}$

A transmission line is said to be lossless if the conductors of the line are perfect ($\sigma = \infty$) & the dielectric medium separating them is lossless ($\sigma = 0$)

characteristic impedance $Z_0 = \sqrt{\frac{L}{C}}$
phase velocity $V_p = \frac{1}{\sqrt{LC}}$

Reflection coefficient

Voltage reflection coefficient is defined as ratio of reflected voltage to the incident voltage at a distance d from receiving end, denoted as Γ (gamma)

$$\Gamma = \frac{V^- e^{-\gamma d}}{V^+ e^{-\gamma d}}$$

At a distance d from receiving end $x = -d$

$$\frac{V^- e^{-\gamma d}}{V^+ e^{-\gamma d}} = \frac{V^-}{V^+} e^{-2\gamma d}$$

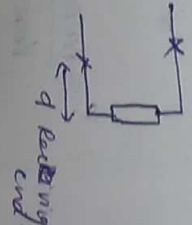
At receiving end or load end, $d=0$

$$\Gamma = \frac{V^-}{V^+} e^0 \quad \Gamma = \frac{V^-}{V^+}$$

i.e. Reflection coeff. at receiving end or load end

$$\boxed{\Gamma = \frac{V^-}{V^+}}$$

Voltage standing wave ratio (VSWR)
line vltg. will show maxima with minimum



Except to min & max and min when line is terminated with a mismatch level.

VSWR is defined as the ratio of max. V to min. V of a fully terminated line.

$$S = \frac{V_{max}}{V_{min}}$$

$$V_{max} = V^+ + V^-$$

$$= V^+ + \sqrt{R} V^+ = V^+ [1 + \sqrt{R}]$$

$$V_{min} = V^+ - V^-$$

$$= V^+ - \sqrt{R} V^+ = V^+ [1 - \sqrt{R}]$$

$$S = \frac{V_{max}}{V_{min}} = \frac{V^+ [1 + \sqrt{R}]}{V^+ [1 - \sqrt{R}]} = \frac{1 + \sqrt{R}}{1 - \sqrt{R}}$$

$$\sqrt{R} = \frac{S-1}{S+1}$$

$$\sqrt{R} = \frac{Z_L - Z_0}{Z_L + Z_0}$$

? A lossless transmission line with characteristic impedance of 75 Ω is terminated by load impedance of 100 Ω . If the magnitude of incident wave is 10V, find the max & min values of V along the

VSWR.

$$\sqrt{R} = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{100 - 75}{100 + 75} = 0.23$$

$$\sqrt{R} = \frac{V_r}{V_i} \Rightarrow V_r = \sqrt{R} V_i$$

$$= 0.23 \times 10 = 2.3V$$

$$V_{max} = V_i + V_r$$

$$= 10 + 2.3 = 12.3V$$

$$V_{min} = V_i - V_r$$

$$= 10 - 2.3 = 7.7V$$

$$VSWR = \frac{V_{max}}{V_{min}} = \frac{12.3}{7.7} = 1.597$$

Characteristic Impedance

General solution of wave equation

$$V(z) = V^+ e^{-\gamma z} + V^- e^{\gamma z}$$

Differentiate above equation

$$\gamma =$$

$$\frac{dV}{dz} = -\gamma V^+ e^{-\gamma z} + \gamma V^- e^{\gamma z}$$

$$0 = (R + j\omega L) I(z) = -\gamma V^+ e^{-\gamma z} + \gamma V^- e^{\gamma z}$$

$$-\gamma(z) = \gamma (-V^+ e^{-\gamma z} + V^- e^{\gamma z})$$

$$\gamma(z) = \frac{R + j\omega L}{R + j\omega L} [V^+ e^{-\gamma z} - V^- e^{\gamma z}]$$

$$= \frac{1}{Z_0} (V^+ e^{-\gamma z} - V^- e^{\gamma z})$$

$$\text{where } Z_0 = \sqrt{\frac{R+j\omega L}{G+j\omega C}}$$

Distortionless line
 Distortionless line is one in which attenuation const. α is frequency independent. while phase const. β is linearly dependent on freq.

$$\gamma = \alpha + j\beta = \sqrt{(R+j\omega L)(G+j\omega C)}$$

for distortionless line $\left[\frac{R}{L} = \frac{G}{C} \right]$

$$\gamma = \sqrt{Rn(1+j\omega L/R)(1+j\omega C/G)} = \alpha + j\beta$$

$$= \sqrt{Rn} (1 + j\frac{\omega L}{n}) = \alpha + j\beta$$

$$\alpha = \sqrt{Rn} \quad \beta = \omega \sqrt{LC}$$

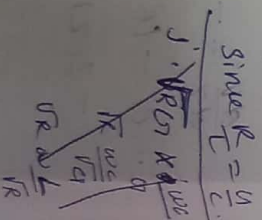
α doesn't depend on freq.

β is linear fun of frequency.

$$Z_0 = \sqrt{\frac{R(1+j\omega L/R)}{G(1+j\omega C/G)}} = \sqrt{\frac{R}{G}} = \sqrt{\frac{L}{C}}$$

$$= R_0 + jX_0$$

$$R_0 = \sqrt{R/G} = \sqrt{\frac{L}{C}} \quad X_0 = 0$$



$$\text{Velocity} = \frac{\omega}{\beta} = \frac{1}{\sqrt{LC}}$$

Phase velocity is independent of freq. b/c of phase const. β linearly depends on ω .

? calculate the VSWR for a transmission line with characteristic impedance of 86Ω & load impedance $Z_L = 155 \Omega$.

$$\overline{R} = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{155 - 86}{155 + 86} = 0.286$$

$$VSWR = \frac{1 + \overline{R}}{1 - \overline{R}} = \frac{1 + 0.286}{1 - 0.286} = 1.801$$

? the 1st constants of a transmission line at 250 MHz is $R = 80 \Omega/\text{m}$, $L = 0.364 \mu\text{H}/\text{m}$, $C = 40 \text{ pF}/\text{m}$ & $G = 71 \text{ mS}/\text{m}$.

calculate Z_0

$$Z_0 = \sqrt{\frac{R+j\omega L}{G+j\omega C}} = \sqrt{\frac{80 + j(2\pi \times 250 \times 10^6 \times 0.364 \times 10^{-6})}{-71 \times 10^{-3} + j(2\pi \times 250 \times 10^6 \times 40 \times 10^{-12})}}$$

$$= \sqrt{9 - j1.263} = \sqrt{9.08 \angle -7.99^\circ}$$

$$= 3.015 \angle -3.995^\circ \Omega$$

? considers a transmission line of $Z_0 = 50 \Omega$, let it be terminated at one end by $j50 \Omega$. find S ?

$$\overline{R} = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{j50 - 50}{j50 + 50} = \frac{j-1}{j+1}$$

$$S = \frac{1 + |\Gamma_R|}{1 - |\Gamma_R|} = \frac{1 + 1}{0} = \infty \quad \left. \vphantom{\frac{1 + |\Gamma_R|}{1 - |\Gamma_R|}} \right\} \text{complete mismatch}$$

? what are the values of load impedance if a purely resistive load when connected to a lossless 75- Ω line produce a VSWR of 3 on the line.

$$Z_0 = 75\text{-}\Omega \quad S = 3 \quad Z_L = ?$$

$$S = \frac{1 + |\Gamma_R|}{1 - |\Gamma_R|}$$

$$\Gamma_R = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$|\Gamma_R| \rightarrow +ve \quad Z_L > Z_0$$

$$3 - 3|\Gamma_R| = 1 + |\Gamma_R|$$

$$|\Gamma_R| \rightarrow -ve \quad Z_L < Z_0$$

$$2 = 4|\Gamma_R|$$

$$|\Gamma_R| = \pm \frac{1}{2}$$

$$\textcircled{1} \Gamma_R = +1/2$$

$$\frac{1}{2} = \frac{Z_L - 75}{Z_L + 75}$$

$$Z_L = 225\text{-}\Omega$$

$$\textcircled{2} \Gamma_R = -1/2$$

$$-1/2 = \frac{Z_L - 75}{Z_L + 75}$$

$$\Rightarrow Z_L = 25\text{-}\Omega$$